Graph Convolutional Network With Relaxed Collaborative Representation for Hyperspectral Image Classification

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***Abstract*— Graph convolutional networks (GCNs) have been skillfully employed in hyperspectral image (HSI) classification, exhibiting remarkable performance owing to their unique superiority in handling non-Euclidean graph-structured data. However, the inherent absence of predefined connections between pixels in HSI results in the underutilization of the structural and attribute information of the graph edges. Furthermore, the construction of adjacency matrices for large-scale HSI data imposes a huge computational burden on traditional GCNs. Therefore, in this article, a novel method combining relaxed collaborative representation (RCR) and GCN (RCR-GCN) for hyperspectral classification is proposed. Specifically, RCR is adopted to compute the representation coefficients of each feature, reflecting the similarity and diversity among different sample features. Meanwhile, the representation coefficients are applied as edge attributes in the graph, denoting the weights of the connections between neighboring nodes. After that, GCN is employed to classify the graph nodes. Moreover, an efficient version of the RCR-GCN method is developed to boost the computation, which constructs the graph based on superpixel nodes instead of the pixel nodes by using simple linear iterative clustering (SLIC). Extensive experiments on three HSI image datasets demonstrate that the proposed method outperforms other state-of-the-art methods and achieves more efficiency and feasibility in HSI image classification.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral classification, relaxed collaborative representation (RCR), simple linear iterative clustering (SLIC), superpixel segmentation.**

I. Introduction

HYPERSPECTRAL image (HSI), characterized by its superior spectral resolution across contiguous narrow bands, has been widely used in precision agriculture [1], [2], mineral identification [3], water quality monitoring [4], etc. In this regard context, HSI classification aims to assign a specific class label to each pixel, which is the cornerstone for these successful applications. Therefore, exploring appropriate methods is of the utmost importance, as well as using limited samples to achieve accurate classification results [5], [6].

Deep learning has provided significant breakthroughs in computer vision [7], [8], [9], [10], [11], [12], [13], [14], which can exploit complex features automatically from massive data without handmade engineering and prior knowledge [15], [16], [17]. Inspired by the success of deep learning, advanced methods are introduced into the field of HSI classification. Among them, convolutional neural network (CNN) is the most widely used due to its high efficiency. Compared with traditional methods, CNN extracts most discriminant feature information from the original data through a series of hierarchical layers. Specifically, the frontier layers extract low-level features and the deeper layers analyze high-level features. The learning process is totally automatic, which makes CNN more suitable for dealing with different kinds of complex situations. In general, different networks extract features favorable to themselves in different ways [18]. Yang et al. [19] first proposed 2-D convolutional neural network (2-D-CNN), 3-D-CNN, recurrent 2-D CNN (R-2-D-CNN) and R-3-D-CNN. The 2-D-CNN can exploit the spatial context, whereas the 3-D-CNN can exploit both the spatial context and the spectral information. They further designed the R-2-D-CNN and the R-3-D-CNN to address the noisy spatial information problem. Lee and Kwon [20] described a deeper and wider CNN called contextual deep CNN, which can optimally explore local contextual interaction by exploiting local spatial–spectral relationship of neighboring pixels. Zhang et al. [21] proposed a classification framework called diverse region-based CNN, which can encode semantic context-aware representation to obtain promising features. He et al. [22] proposed MCMs-2DCNN model based on multiscale covariance maps (MCMs), which increases the robustness of the CNN. Paoletti et al. [23] developed a 3-D-CNN that implements a border mirroring strategy to effectively process border areas in the image.

Although CNN has achieved great performance in HSI classification [24], it merely actives in Euclidean grid data rather than non-Euclidean data. In this regard, graphs are a generalization of grid data and can be more expressive

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for ground objects in HSI. It has many applications in social networks [25], [26], knowledge graphs [27], [28], and chemical molecules [29], [30], based on node classification, edge prediction, and graph identification. The emergence of graph convolutional network (GCN) achieves deploying deep learning in graphs and builds a bridge between deep learning and non-Euclidean data. In particular, GCN can be used to model long-range spatial relations of pixels in HSI, which generally fails to be considered in CNN. Presently, there are a large number of applications of GCN in HSI classification. Qin et al. [31] proposed a spectral–spatial GCN (S^2GCN) method for semisupervised classification which considers both adjacent nodes in graph and neighboring pixels in spatial domain. Mou et al. [32] proposed a graph-based semisupervised network called nonlocal GCN, which takes both labeled and unlabeled data into use. Wan et al. [33] proposed a multiscale dynamic GCN (MDGCN) utilizing a dynamic graph convolution operation, where the similarity measures among pixels can be updated by fusing current feature embeddings, and the graph can be gradually refined during the convolution process of GCN, making the feature embeddings more accurate. Later, they proposed a context-aware dynamic GCN (CAD-GCN) [34], in which the MDGCN is employed as the backbone and the regions of CAD-GCN are adaptively learned by projection and reprojection steps. At the same time, the graph edges connecting different regions are updated properly. Hong et al. [35] developed a minibatch GCN (miniGCN), which trains a large-scale GCN in a minibatch fashion and is capable of inferring out-of-sample data without retraining networks.

Depending on the graph nodes construction, GCN can be divided into pixel-based and superpixel-based methods. Regarding superpixel-based methods, they split HSI into superpixels in which pixels are similar to each other and adjacent in space to construct a graph using fewer nodes. Since constructing a graph on all pixels would consume almost unbearable resources owing to the huge number of nodes and edges in HSI, superpixel segmentation usually performs as a preprocessing step to improve efficiency. Gong et al. [35] proposed a superpixel spectral–spatial feature fusion GCN (S3FGCN), which can comprehensively use information in spectral, spatial, and spectral–spatial domains with limited data. Zhang et al. [37] proposed an end-to-end mixhop superpixel-based GCN (EMS-GCN) framework in which a superpixel graph is constructed and both the local information within superpixels and long-range information among superpixels are extracted. Yang et al. [38] proposed a dynamic multiscale GCN classifier (DMSGer), which regards over-segmented multiple homogeneous regions (superpixels) as the nodes to construct a relatively small graph, and a new pixel-level graph nodes updating (PGNU) strategy is presented then regions can be fully explored simultaneously. Ding et al. [39] developed a graph convolution with adaptive filters and aggregator fusion (AF2GNN), in which superpixel segmentation is employed to refine the local spatial features and a two-layer 1-D-CNN is proposed to transform the spectral features of superpixels. Jia et al. [40] proposed a graph-in-GCN (GiGCN) which is composed of internal and external graph convolution to extract hierarchical features from the superpixel viewpoint. The GiG covers information inside and outside superpixels corresponding to local and global features.

Different from deep learning, representation learning is a parameter-free method which can use the original spectral features directly to classify the testing samples. Specifically, this type of methods constructs a dictionary by selecting samples from each class, and then, focusing on how to represent the similarity between dictionary and testing samples, assign the classes labels of testing samples based on representation residuals. The original representation method mainly includes sparse representation (SR) [41], [42] and collaborative representation (CR) [43]. The SR uses the l_1 norm, and the CR uses the l_2 norm with lower complexity but similar performance. However, these two methods only focus on calculating the similarity between samples but ignore the diversity between features, which are an important natural characteristic and has a significant effect on the representation. Hence, relaxed CR (RCR) was proposed [44] to improve the performance. In the RCR, each feature is coded on its dictionary. The variance of coding coefficients is required to be minimized to utilize the similarity between features, and the distance between features is interpreted as the reference to compute the coefficients representing diversity. Several algorithms have been developed to demonstrate the validity of RCR in HSI classification. Su et al. [45] proposed a RCR with a band-weighting framework based on superpixel segmentation, where multifeatures are further put into service to improve the classification performance. Gao et al. [46] introduced dictionary learning into the RCR, and two new indicators formed by optimizing coding coefficients were designed to measure the discriminability. Liu et al. [47] presented a region-based relaxed multiple kernel CR method which fused the spatial multiscale features and spectral features in the framework. Dundar et al. [48] proposed a method called GRR which combines guided filter (GF) of a random patches network (RPNet) with the RCR.

In the current study, although the GCN performed well in HSI classification, the structure and properties of edges in graph have not been fully utilized because there is no prior connection between pixels in HSI. When constructing the graph, the edge is generally roughly assigned a value of 0 or 1, which only indicates whether the adjacent nodes have the connection, ignoring the influence of the connection relationship closeness on the classification.

In this article, a new method combining the RCR and the GCN is proposed for HSI classification the so-called "RCR-GCN." Primarily, the representation coefficients calculated by the RCR are exploited as the edge attribute of the graph, owing to the fact that they reflect the similarity and diversity between different features. Specifically, the representation coefficient is used as the weight of the corresponding edge, and then GCN is used to classify the graph nodes. Besides, the traditional GCN needs to construct an adjacency matrix, which brings a huge computational burden as the HSI data scale is large. In this case, the simple linear iterative clustering (SLIC) [49], [50] method is further employed to reduce the amount of calculation, and the graph is performed

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on superpixel nodes rather than pixel nodes. Thus, the cost-effective method SRCR-GCN is developed. The main contributions are summarized as follows.

1) First of all, our method combines the benefits of representation learning with the advantages of GCN. Representation learning is parameter-free, and the GCN can be used into non-Euclidean data, as well as process the long-range spatial information of pixels in HSI.
2) In order to make full use of the structure and properties of the graph edges, representation coefficient is calculated by the RCR as the weight of edge, indicating connection tightness between adjacent nodes.
3) Considering the high complexity of calculating the coefficients and the huge computational cost of constructing the adjacency matrix, the superpixel segmentation method is employed as a preprocessing step and each superpixel is embedded as a node in the graph. Meanwhile, the superpixel blocks can characterize the local spatial information to facilitate classification.

The rest of the article is organized as follows. The related work, including RCR and GCN, is briefly introduced in Section II. Section III introduces the proposed methods in detail, that is, an original RCR-GCN and the superpixel-based version. The experimental results and analysis to evaluate the two proposed methods are given in Section IV. Finally, conclusions and future work are presented in Section V.

II. Related Work

A. RCR

In general, a sample $\boldsymbol{y}$ is linearly represented by the corresponding dictionary $\mathbf{D}$ with a coefficient $\boldsymbol{a}$ following the linear regression problem:

$$\arg\min_{\boldsymbol{a}} \|\boldsymbol{y} - \mathbf{D}\boldsymbol{a}\|_2^2 + \lambda \|\boldsymbol{a}\|_{l_p} \tag{1}$$

when the regularization constraint $\boldsymbol{p} = 1$, it behaviors a SR model, whereas it is a CR model when $\boldsymbol{p} = 2$.

Equation (1) shows that both the SR and the CR assume all attributes features of training samples to share the same coding vector, which ignores the similarities and diversities between the different features of the model. To this end, Yang et al. [44] proposed the RCR, which can build a model from each feature dimension. During the computing stage, the regularization constraint on the representation coefficients is imposed from each feature, while minimizing the difference between all coefficients and the mean of the coefficients. The weight is optimized according to the coefficient to measure the diversity between features. The objective function of the RCR is

$$\arg\min_{\boldsymbol{a}_b} \sum_{b=1}^{B} \left\|\boldsymbol{y}_b - \mathbf{D}_b\boldsymbol{a}_b\right\|_2^2 + \lambda \|\boldsymbol{a}_b\|_2^2 + \tau \boldsymbol{w}_b \left\|\boldsymbol{a}_b - \bar{\boldsymbol{a}}\right\|_2^2 \tag{2}$$

where $\boldsymbol{B}$ is the number of features, and λ and τ are scale parameters. The mean of coefficients can be calculated as

$$\bar{\boldsymbol{a}} = \boldsymbol{w}_b\boldsymbol{a}_b \Big/ \sum_{b=1}^{B} \boldsymbol{w}_b. \tag{3}$$

If $\boldsymbol{w}_b$ is known, (3) can have a closed-form solution. Due to the weakness of the prior information, the regularization of entropy is applied as the third additional coefficient constraint term of the objective function (2). Specifically, the constraint function is

$$\min_{\boldsymbol{w}_b} \sum_{b=1}^{B} \tau \boldsymbol{w}_b \left\|\boldsymbol{a}_b - \bar{\boldsymbol{a}}\right\|_2^2 + \gamma \boldsymbol{w}_b \ln \boldsymbol{w}_b, \tag{4}$$

where γ is a Lagrange multiplier. Then, weights can be updated by

$$\boldsymbol{w}_b = \exp\left(-1 - \tau \left\|\boldsymbol{a}_b - \bar{\boldsymbol{a}}\right\|_2^2 \Big/ \gamma\right). \tag{5}$$

B. GCN

In graph structure, the graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ is a set with $\boldsymbol{n}$ nodes. If the node $\boldsymbol{v}_i$ connects to the node $\boldsymbol{v}_j$, then $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ are neighbors of each other. The neighbor set of $\boldsymbol{v}_i$ is denoted as $\mathcal{N}_{v_i}$. $\mathcal{E} = \{(\boldsymbol{v}_i, \boldsymbol{v}_j)\}$ is an edge set in which $\boldsymbol{v}_i$ connects to $\boldsymbol{v}_j$. According to the connection relationship, the adjacency matrix $\mathbf{A}$ is

$$\mathbf{A}_{ij} = \begin{cases} 1, & (\boldsymbol{v}_i, \boldsymbol{v}_j) \in \mathcal{E} \\ 0, & (\boldsymbol{v}_i, \boldsymbol{v}_j) \notin \mathcal{E}. \end{cases} \tag{6}$$

The degree matrix $\mathbf{D}_\mathbf{e}$ of the graph $\mathcal{G}$ is a $\boldsymbol{n} \times \boldsymbol{n}$ diagonal matrix, and the elements on the diagonal are the degrees corresponding to the nodes

$$\mathbf{d}_{i,j} = \begin{cases} \mathbf{d}(\boldsymbol{v}_i), & \text{if } \boldsymbol{i} = \boldsymbol{j} \\ 0, & \text{else.} \end{cases} \tag{7}$$

If the edges in a graph not only represent the connections but also have weights that represent the strength and weakness of the connections, the graph is defined as a weighted graph. In the weighted graph, the elements of the adjacency matrix are no longer just [0, 1], but can be an arbitrary positive number. The degree also changes to the sum of the weights of the edges connected to the node. Since the weights of the nonadjacent nodes are 0, the degree is also equal to the sum of the elements of the corresponding row in the adjacency matrix $\mathbf{A}$

$$\boldsymbol{d}(\boldsymbol{v}_i) = \sum_{\boldsymbol{v}_j \in \mathcal{N}_{v_i}} \mathbf{A}_{ij} = \sum_{\boldsymbol{v}_j \in \mathcal{V}} \mathbf{A}_{ij}. \tag{8}$$

The Laplacian matrix $\mathbf{L} = \mathbf{D}_\mathbf{e} - \mathbf{A}$ contains the structure information of the graph. To enhance the generalization ability of the graph, the symmetric normalized Laplacian matrix can be used as $\tilde{\mathbf{L}} = \mathbf{D}_\mathbf{e}^{-1/2}\mathbf{L}\mathbf{D}_\mathbf{e}^{-1/2}$.

In graph Fourier, $\tilde{\mathbf{L}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T$, where $\mathbf{U}$ is an orthogonal matrix, $\mathbf{U} = [\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_n]$. Besides, $\boldsymbol{\Lambda} = \boldsymbol{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is a diagonal matrix, each λ_i of which corresponds to an eigenvector $\boldsymbol{u}_i$. In this sense, each $\boldsymbol{u}_i$ in $\mathbf{U}$ is equivalent to a Fourier base, while each λ_i is equivalent to the frequency that attaches to the corresponding base $\boldsymbol{u}_i$. Hence, Fourier transformation of a graph signal $\boldsymbol{x}$ is $\tilde{\boldsymbol{x}} = \mathbf{U}^T\boldsymbol{x}$ and its inversing operation is $\boldsymbol{x} = \mathbf{U}\tilde{\boldsymbol{x}}$, where $\boldsymbol{x}$ denotes the feature vector of all nodes. The convolution operation of the graph signal $\boldsymbol{x}$ and filter $\boldsymbol{f}$ can be expressed as

$$\boldsymbol{x} * \boldsymbol{f} = \mathbf{U}\left(\mathbf{U}^T\boldsymbol{x} \odot \mathbf{U}^T\boldsymbol{f}\right) \tag{9}$$

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where $\odot$ represents the Hadamard product. According to this formula, if considering the $\mathbf{U}^T \boldsymbol{f}$ as a parameterizable convolution kernel θ, then

$$\begin{aligned} \boldsymbol{x} * \boldsymbol{f} &= \mathbf{U}\left(\mathbf{U}^T \boldsymbol{x} \odot \theta\right) \\ &= \mathbf{U} \boldsymbol{f}_\theta \mathbf{U}^T \boldsymbol{x} \end{aligned} \tag{10}$$

$$\boldsymbol{f}_\theta = \boldsymbol{diag}(\theta) = \begin{pmatrix} \theta_1 & & & \\ & \theta_2 & & \\ & & \ddots & \\ & & & \theta_n \end{pmatrix}. \tag{11}$$

Kipf and Welling [51] used one-order Chebyshev polynomial to simplify the graph convolution as

$$\begin{aligned} \boldsymbol{x} * \boldsymbol{f} &= \theta_0 \boldsymbol{x} + \theta_1 \left(\tilde{\mathbf{L}} - \mathbf{I}_n\right) \boldsymbol{x} \\ &= \theta_0 \boldsymbol{x} - \theta_1 \mathbf{D}_{\mathbf{e}}^{-1/2} \mathbf{A} \mathbf{D}_{\mathbf{e}}^{-1/2} \boldsymbol{x}. \end{aligned} \tag{12}$$

To further reduce the number of parameters and prevent overfitting, $\theta = \theta_0 = -\theta_1$ is set, so that

$$\boldsymbol{x} * \boldsymbol{f} = \theta \left(\mathbf{I}_n + \mathbf{D}_{\mathbf{e}}^{-1/2} \mathbf{A} \mathbf{D}_{\mathbf{e}}^{-1/2}\right) \boldsymbol{x}. \tag{13}$$

The eigenvalues range of $\mathrm{I}_n + \mathbf{D}_{\mathbf{e}}^{-1/2} \mathbf{A} \mathbf{D}_{\mathbf{e}}^{-1/2}$ is [0, 2], multiple iterations would lead to the problems of numerical instability and gradient explosion/vanishing. In order to alleviate this problem, it is necessary to do another normalization. By defining $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_n$, the normalization of the matrix becomes $\tilde{\mathbf{D}}_{\mathbf{e}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}_{\mathbf{e}}^{-1/2}$. In practice, multilayer graph convolutions are often used, with $\mathbf{H}^l$ representing the lth node vector, the $\mathbf{W}^l$ representing the parameters and $\hat{\mathbf{A}} = \tilde{\mathbf{D}}_{\mathbf{e}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}_{\mathbf{e}}^{-1/2}$, the graph convolution of each layer is formed as

$$\mathbf{H}^{l+1} = f\left(\mathbf{H}^l, \mathbf{A}\right) = \sigma\left(\hat{\mathbf{A}} \mathbf{H}^l \mathbf{W}^l\right). \tag{14}$$

III. Proposed Method

One of the problems when applying graph neural networks to hyperspectral data is that the structure and property of the edges cannot be fully utilized because there is no prior connection between pixels. Therefore, this article proposes a new method to use the representation coefficient as the edge attribute of graph structure data, that is, the representation coefficient is used as the weight of the edge to obtain the weighted graph, and then the GCN is used to classify the graph nodes.

A. RCR-GCN

A simple trial of the CR-GCN is performed first. Fig. 1 shows the process of obtaining the CR representation coefficient and using it as the edge weight. The expression of the CR is

$$\arg\min_{\boldsymbol{a}} \|\boldsymbol{y} - \mathbf{D}\boldsymbol{a}\|_2^2 + \lambda \|\boldsymbol{a}\|_2^2 \tag{15}$$

and the representation coefficient $\boldsymbol{a}$ can be obtained as

$$\boldsymbol{a} = \left(\mathbf{D}^T \mathbf{D} + \lambda \mathbf{I}\right)^{-1} \left(\mathbf{D}^T \boldsymbol{y}\right). \tag{16}$$

Although all subdictionaries are included in the representation, the CR still does not take full advantage of different features. However, it is important to balance the structural similarity and feature diversity of the representation coefficients. The RCR assigns a representation coefficient to each feature, and the obtained $\boldsymbol{a}$ is consistent in structure and well reflects the diversity between different features in the representation.

Based on the above, the RCR-GCN is proposed, Fig. 2 shows the overall architecture in detail and Algorithm 1 summarizes the steps. First, the ith testing sample $\boldsymbol{y}^i$ is calculated by using the RCR method, and the corresponding representation coefficient is $\boldsymbol{a}_b^i$, $\forall \boldsymbol{a}_b^i \in \{\boldsymbol{a}_1^i, \boldsymbol{a}_2^i, \ldots, \boldsymbol{a}_B^i\}$. In the GCN, each $\boldsymbol{a}_b^i$ is used as the weight of the edge to represent the connection between the dictionary $\mathbf{D}$ and the testing sample $\boldsymbol{y}^i$, indicating how tightly the connection is. Then the objective function of the RCR is

$$\arg\min_{\boldsymbol{a}_b^i} \sum_{b=1}^{B} \left\|\boldsymbol{y}_b^i - \mathbf{D}_b \boldsymbol{a}_b^i\right\|_2^2 + \lambda \left\|\boldsymbol{a}_b^i\right\|_2^2 + \tau \boldsymbol{w}_b^i \left\|\boldsymbol{a}_b^i - \bar{\boldsymbol{a}}^i\right\|_2^2. \tag{17}$$

In the CR, $\boldsymbol{A}$ is an $\boldsymbol{n} \times \boldsymbol{n}$ matrix, which can be sparsely represented to save memory, where $\boldsymbol{n}$ is the total number of samples. For $\boldsymbol{A}_{i,j}$, $\boldsymbol{i}$ is the number of subscript for testing sample, $\boldsymbol{j}$ is the subscript for subdictionary, and $\boldsymbol{A}_{i,j} = \boldsymbol{a}^{i,j}$. In the RCR, $\boldsymbol{A}$ is an $\boldsymbol{n} \times \boldsymbol{n} \times \boldsymbol{B}$ matrix, and $\boldsymbol{A}_{i,j,:} = \boldsymbol{a}_b^{i,j}$. Fig. 3 illustrates a CR-GCN adjacency matrix, while Fig. 4 shows the structure of RCR-GCN adjacency matrix.

Algorithm 1 RCR-GCN Algorithm

Input:

(1) Hyperspectral image features
(2) Set RCR parameters: regulation parameter λ, balance parameter τ, Lagrange multiplier γ
(3) Set GCN parameters

Main Loop:

1. Input: dictionary $\mathbf{D}$ and testing set $\boldsymbol{y}$
2. Initialization: $\boldsymbol{w}_b = 1$
3. RCR
   for each testing sample $\boldsymbol{y}^i$
   while $\left\|\boldsymbol{w}_b^{t+1} - \boldsymbol{w}_b^t\right\|_2 / \left\|\boldsymbol{w}_b^t\right\|_2 > \zeta$ (t represents t^{th} iteration and ζ is a constant)
   Update coefficient vectors $\boldsymbol{a}_b^i$
   Update weights $\boldsymbol{w}_b^i$
   end while
   end for
4. Update the adjacency matrix A
5. GCN
   Predict via (18)

Output: $\hat{\boldsymbol{Y}}$

Then, a two-layer GCN is used to classify graph nodes, and the node attribute matrix in the graph is $\mathbf{X} \in \mathbb{R}^{n \times d}$. Each node in the graph can be divided into one class. The following method is used to predict the labels of nodes:

$$\begin{aligned} \hat{\mathbf{Y}} &= f(\mathbf{X}, \mathbf{A}) \\ &= \mathrm{Softmax}\left(\hat{\mathbf{A}} \mathrm{ReLU}\left(\hat{\mathbf{A}} \mathbf{X} \mathbf{W}^0\right) \mathbf{W}^1\right). \end{aligned} \tag{18}$$

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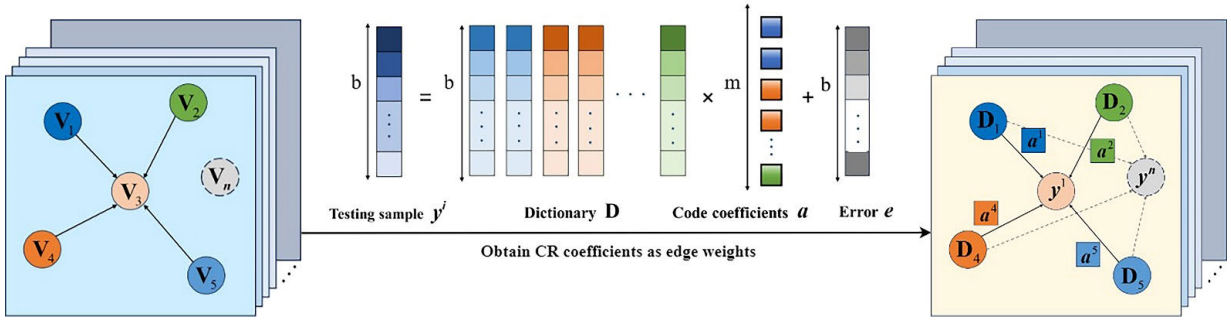


Fig. 1. Illustration of CR-GCN. Use CR representation coefficients as the weight of edges to construct the graph.

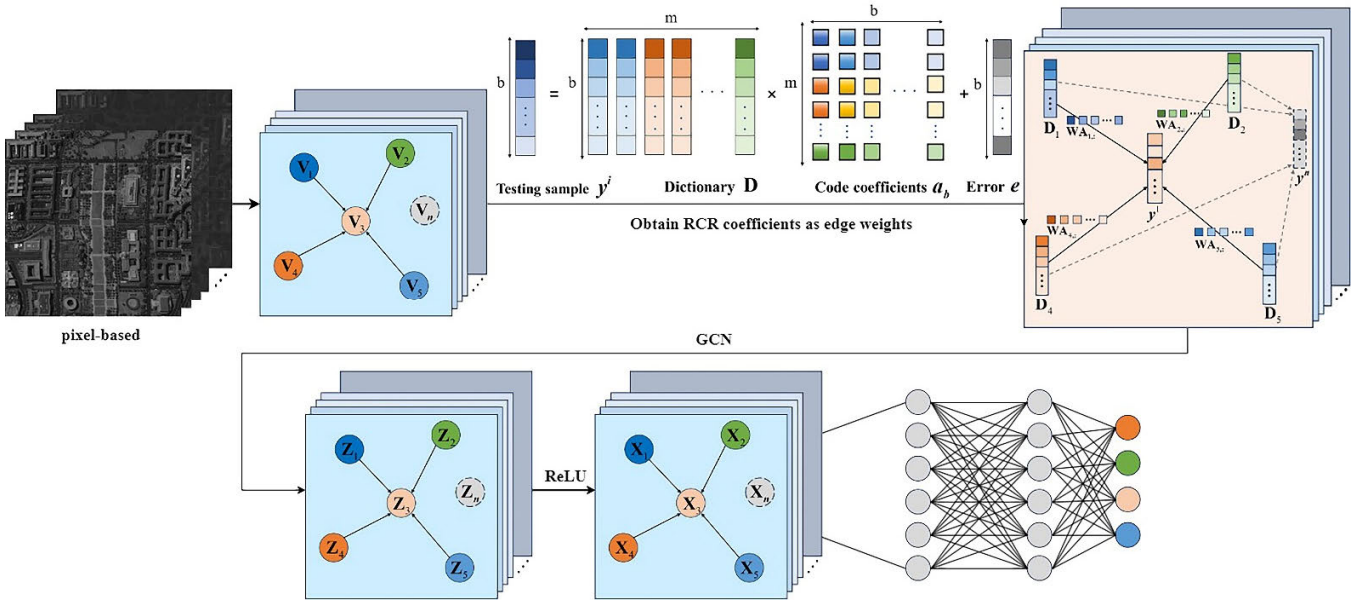


Fig. 2. Overall architecture of RCR-GCN.

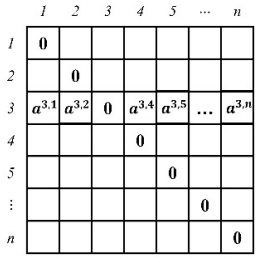


Fig. 3. Adjacency matrix of CR.

First of all, the node attribute matrix X and the adjacency matrix A of the whole graph are input into the convolutional network to obtain the node embedding matrix, and then the Softmax is adopted to generate the predict classification results. Finally, compare the gap between the predicted results and the real label in the training set to obtain the loss

$$\mathcal{L} = -\sum_{l=0}^{m-1} \sum_{i \in \mathcal{V}_{train}} \mathbf{Y}_{li} \ln \hat{\mathbf{Y}}_{li} \tag{19}$$

where $\boldsymbol{m}$ represents the total number of classes. The weights of this network can be obtained by using the stochastic gradient descent method, and then the testing samples are classified.

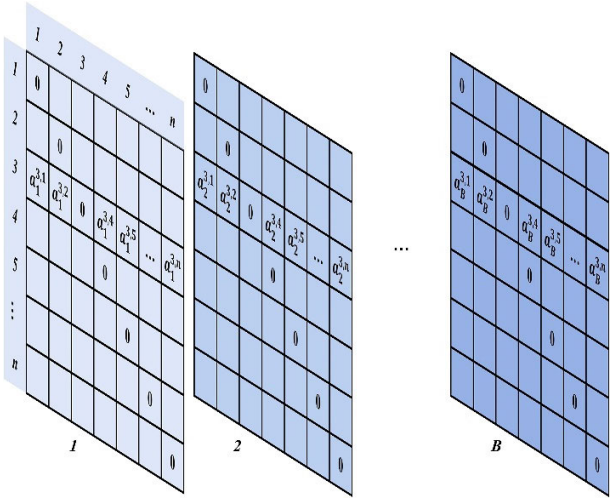


Fig. 4. Adjacency matrix of RCR.

B. SRCR-GCN

The SLIC is an image segmentation method which can generate compact and nearly uniform superpixels, possessing a high comprehensive evaluation in terms of computing speed, object contour preservation, and superpixel shape.

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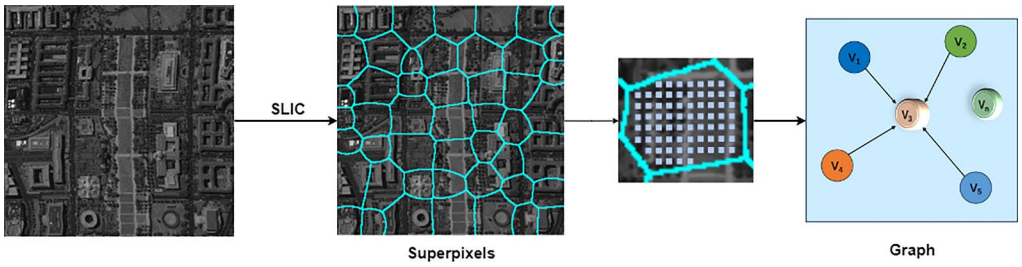


Fig. 5. Illustration of superpixel-based GCN. Use SLIC as the image segmentation method.

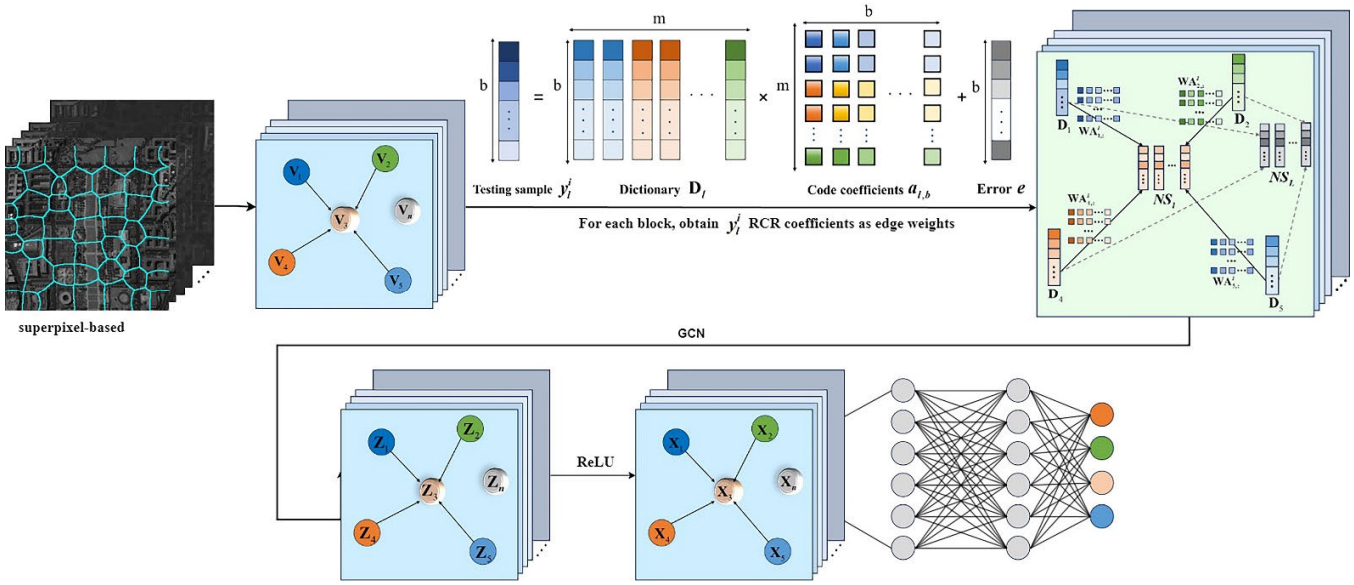


Fig. 6. Overall architecture of SRCR-GCN.

Since the HSI contains innumerable pixels, the computational complexity for subsequent graph convolution and classification may be unacceptable. To alleviate this problem, the SLIC is adopted to segment the entire image into superpixels, where each superpixel represents a region with high spectral-spatial similarity. Then, each superpixel instead of the pixel is treated as a graph node. Therefore, the amount of graph nodes can be significantly reduced, and the computational efficiency can be improved. Fig. 5 is the illustration of superpixel-based GCN, where each node in the graph is a superpixel block. Besides, the generated superpixels is helpful for classification, as the local structural information is preserved and nearby pixels with high spatial consistency are more likely to belong to the same class.

To reduce the amount of calculation, the graph is usually performed on superpixel nodes rather than pixel nodes. In this section, the SRCR-GCN method based on superpixel segmentation is presented to improve the accuracy and stability through optimizing representation coefficients and weights simultaneously. Moreover, principal components analysis (PCA) is used to reduce the dimension of HSI before using SLIC. Fig. 6 depicts the overall architecture of SRCR-GCN.

Superpixel blocks $\mathbf{NS}_l (l = 1, 2, \ldots, L)$ obtained from SLIC are used to form testing sample sets to ensure uniform distribution. For each block, use the preselected subdictionary $\mathbf{D}_l, \forall \mathbf{D}_l \in \{\mathbf{D}_1, \mathbf{D}_2, \ldots, \mathbf{D}_L\}$. Then, the proposed RCR-GCN is applied in each block for classification. Although the features in a block merely retain the spectral information for arraying as vectors, the superpixels have retained the original spatial information.

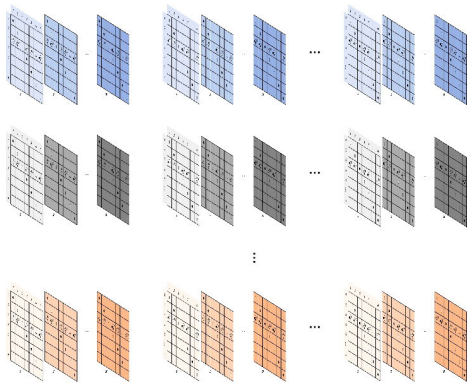


Fig. 7. Adjacency matrix of SRCR.

The objective function of SRCR-BW is shown as

$$\arg\min_{a_{l,b}^i} \sum_{b=1}^{B} \left\| y_{l,b}^i - \mathbf{D}_{l,b} a_{l,b}^i \right\|_2^2 + \lambda \left\| a_{l,b}^i \right\|_2^2 + \tau w_{l,b} \left\| a_{l,b}^i - \bar{a}_l \right\|_2^2. \quad (20)$$

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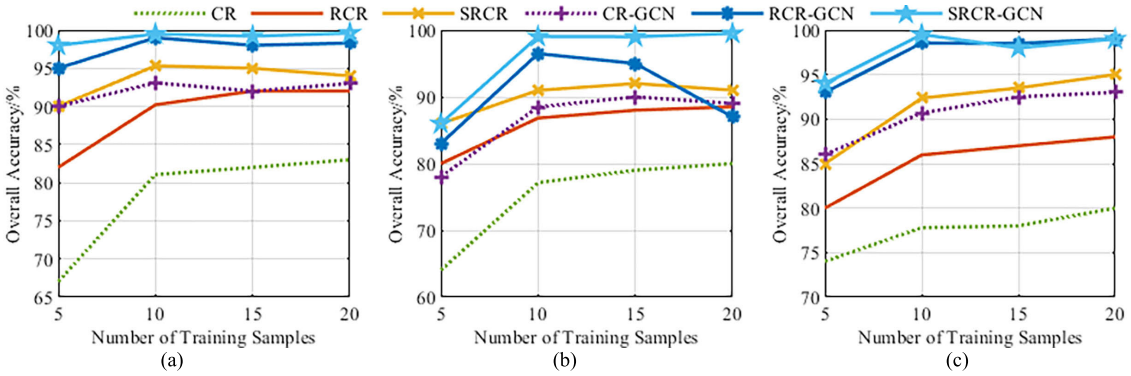


Fig. 8. Results of OA in different datasets. (a) dc. (b) PD. (c) YC.

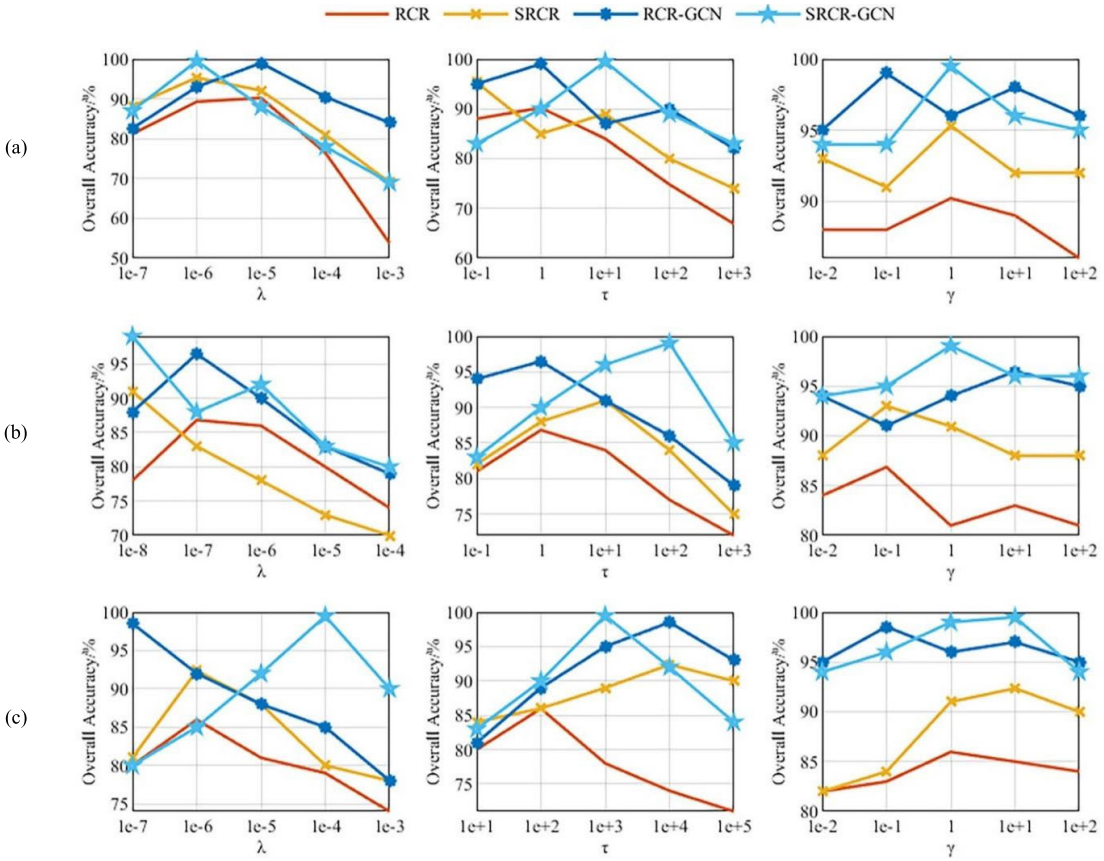


Fig. 9. Results of λ, τ, and γ parameters analysis in different datasets. (a) dc. (b) PD. (c) YC.

A closed-form solution can be calculated as

$$\boldsymbol{a}_{l,b}^{i} = \boldsymbol{a}_{l,b}^{0} + \tau \frac{\boldsymbol{w}_{l,b}}{\sum_{\eta=1}^{B} \boldsymbol{w}_{l,\eta}} \mathbf{U}_{l,b} \mathbf{V}_{l} \sum_{\eta=1}^{B} \boldsymbol{w}_{l,\eta} \boldsymbol{a}_{l,\eta}^{0} \tag{21}$$

with

$$\mathbf{U}_{l,b} = \left(\mathbf{D}_{l,b}^{T} \mathbf{D}_{l,b} + \mathbf{I}\left(\lambda + \tau \boldsymbol{w}_{l,b}\right)\right)^{-1} \tag{22.1}$$

$$\boldsymbol{a}_{l,b}^{0} = \mathbf{U}_{l,b} \mathbf{D}_{l,b}^{T} \boldsymbol{y}_{l,b} \tag{22.2}$$

$$\mathbf{V}_{l} = \left(\mathbf{I} - \sum_{\eta=1}^{B} \tau \mathbf{U}_{l,\eta} / \boldsymbol{B}\right)^{-1} \tag{22.3}$$

$$\bar{\boldsymbol{a}}_{l} = \sum_{b=1}^{B} \boldsymbol{w}_{l,b} \boldsymbol{a}_{l,b} \Big/ \sum_{b=1}^{B} \boldsymbol{w}_{l,b}\,. \tag{22.4}$$

Due to the lack of prior information, the objective function is reduced to

$$\min_{\boldsymbol{w}_{l,b}} \sum_{b=1}^{B} \tau \boldsymbol{w}_{l,b} \left\| \boldsymbol{a}_{l,b} - \bar{\boldsymbol{a}}_{l} \right\|_{2}^{2} + \gamma \sum_{b=1}^{B} \boldsymbol{w}_{l,b} \ln \boldsymbol{w}_{l,b}. \tag{23}$$

Assume the initial weight $\boldsymbol{w}_{b}^{0} = 1$, then, update the weights in the $\boldsymbol{l}$th superpixel block with

$$\boldsymbol{w}_{b} = \exp\left(-1 - \tau \left\| \boldsymbol{a}_{l,b} - \bar{\boldsymbol{a}}_{l} \right\|_{2}^{2} / \gamma\right). \tag{24}$$

Fig. 7 shows the structure of SRCR-GCN adjacency matrix. After traversing all blocks, it produces a reconstructed image. Algorithm 2 summarizes the steps.

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Algorithm 2 SRCR-GCN Algorithm

Input:

(1) Hyperspectral image features
(2) Set RCR parameters: regulation parameter λ, balance parameter τ, Lagrange multiplier γ, number of superpixels ($\boldsymbol{L}$)
(3) Set GCN parameters
(4) Segmentation: generate the blocks **NS** using SLIC

Main Loop:

For each block $\mathbf{NS}_l$

1. Input: sub-dictionary $\mathbf{D}_l$ and testing set $\mathbf{Y}_l$ ($\boldsymbol{l}$ is the number of superpixel block)
2. Initialization: $\boldsymbol{w_b} = 1$
3. SRCR

for each testing sample $\boldsymbol{y}_l^i$ in the $\boldsymbol{l}^{th}$ block
while $\left\|\boldsymbol{w}_b^{t+1} - \boldsymbol{w}_b^t\right\|_2 / \left\|\boldsymbol{w}_b^t\right\|_2 > \zeta$ ($\boldsymbol{t}$ represents $\boldsymbol{t}^{th}$ iteration and ζ is a constant)
Update coefficient vectors $\boldsymbol{a}_{l,b}^i$
Update weights $\boldsymbol{w}_{l,b}$
end while
Update the adjacency matrix $\mathbf{A}_l^i$
predict $\hat{\mathbf{Y}}_l^i$ via GCN
end for

End for

Output: $\hat{\mathbf{Y}}$

IV. Experiments

A. Dataset

1) Mall of Washington DC (DC): The first dataset is acquired by the HYDICE sensor which includes 210 bands ranging from 400 to 2400 nm. Due to the complex atmospheric environment, 191 bands are used for experiments. It has 266 × 304 pixels, and there are six classes including road, grass, trees, and so on. Table I gives the detail of labeled samples.

2) Purdue Campus (PD): The second dataset is collected by the HyMap system which includes 126 bands, and all of them are used for experiments. It has 377 × 512 pixels, and there are six classes including roof, soil, grass, and so on. Table II gives the detail of labeled samples.

3) Dafeng Natural Reserve (Yancheng Wetland, YC): The third experimental dataset is acquired by GF-5, which includes 256 bands, and all of them are used for experiments. It has 986 × 632 pixels and nine different classes including sea, mud flat, river, and so on. As a typical coastal wetland, it is located in Yancheng, Jiangsu, China. The detailed information of the labeled samples is shown in Table III.

B. Classification

1) Experimental Setup: Since the proposed methods are mainly about an improvement of the representation and the GCN methods, some related methods, such as CRC, CNN, GAT, GiGCN, and miniGCN are used for comparison in order to illustrate the effectiveness of RCR combined with GCN. Specifically, GCN is compared with CNN and GAT to illustrate the effectiveness of graph convolution. GiGCN and miniGCN are the latest GCN based models that GiGCN can extract both local and global features from the superpixel viewpoint and miniGCN trains large-scale data in a minibatch fashion. SRCR and SRCR-GCN are compared with RCR and RCR-GCN to illustrate the effectiveness of segmentation. The number of training samples for each class is set to [5, 10, 15, and 20], and the remaining samples are considered as testing set. The regularization parameter λ is set to [1e-7, 1e-6, 1e-5, 1e-4, 1e-3] for dc and YC dataset, and [1e-8, 1e-7, 1e-6, 1e-5, 1e-4] for PD dataset. The balance parameter τ is set to [1e-1, 1, 1e+1, 1e+2, 1e+3] for dc and PD dataset, and [1e+1, 1e+2, 1e+3, 1e+4, 1e+5] for YC dataset. The Lagrange multiplier γ is set to [1e-2, 1e-1, 1, 1e+1, 1e+2] for all related algorithms. The number of iterations is five according to the speed of convergence. Besides, the epoch is set to 100, and the learning rate is 0.001 in the network. Three indicators, including overall accuracy (OA), average accuracy (AA), and kappa coefficients, are calculated to evaluate the effectiveness.

2) Parameters Analysis: Fig. 8 shows the changes in classification accuracy as the number of training samples increases. It can prove that the accuracy of the methods increase slightly as the number of training samples increases, and the high accuracy of SRCR-GCN method is stable.

TABLE I
SIX GROUND-TRUTH CLASSES OF THE HYDICE DATASET

No	Class	Samples
C1	Road	947
C2	Grass	963
C3	Shadow	589
C4	Trail	624
C5	Tree	679
C6	Roof	1139
Total		4941

TABLE II
SIX GROUND-TRUTH CLASSES OF THE HYMAP DATASET

No	Class	Samples
C1	Road	1287
C2	Grass	1114
C3	Shadow	219
C4	Soil	379
C5	Tree	1351
C6	Roof	1285
Total		5635

TABLE III
NINE GROUND-TRUTH CLASSES OF THE GF-5 DATASET

No	Class	Samples
C1	Sea	1020
C2	Mud flat	528
C3	Road	150
C4	Building	89
C5	Farmland	918
C6	River	276
C7	Tree	724
C8	Aquaculture	746
C9	Lake	79
Total		4530

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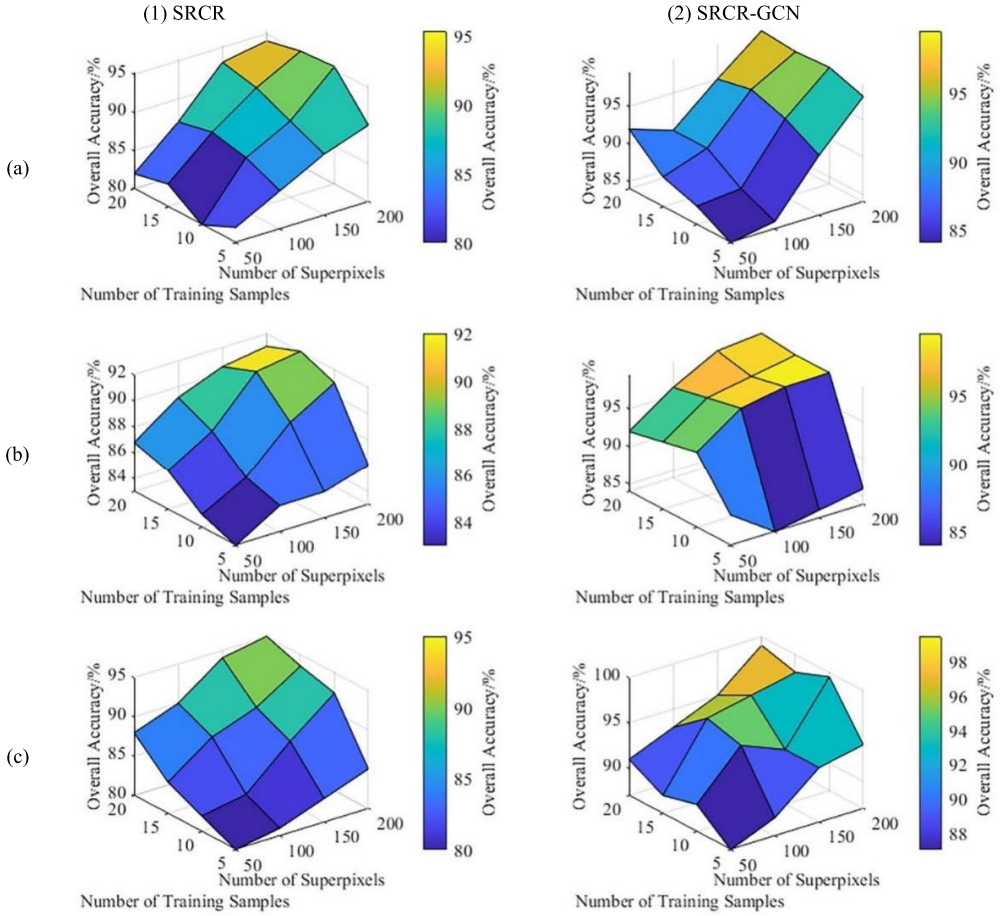


Fig. 10. OA results of SRCR and SRCR-GCN methods, set different number of training samples and superpixel blocks in datasets. (a) dc. (b) PD. (c) YC.

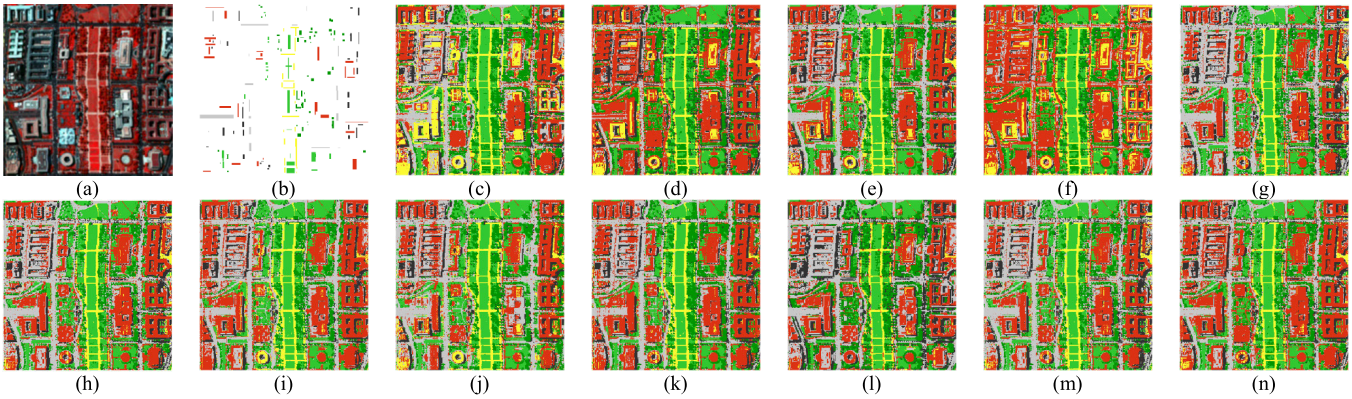


Fig. 11. Classification results of all samples from dc dataset. (a) False-color image. (b) Ground truth. (c) CR. (d) RCR. (e) CNN. (f) GAT. (g) GiGCN. (h) miniGCN. (i) GCN. (j) CR-GCN. (k) RCR-GCN. (l) SRCR. (m) SGCN. (n) SRCR-GCN.

The effects of the proposed methods on the three parameters are evaluated, as shown in Fig. 9. In general, the accuracy is sensitive to λ and τ. For dc dataset, the classification accuracy first increases as the value of λ and τ increase, and then reaches its best. They are more sensitive for the SRCR-GCN method than the RCR-GCN. For PD dataset, as the value of λ and τ increases, the classification accuracy decreases steadily. Similarly, SRCR-GCN is more sensitive to changes in parameters than RCR-GCN. For YC dataset, the situation varies that as the value of λ increases, the accuracy of RCR-GCN decreases but the accuracy of SRCR-GCN increases and reaches its best. Besides, the accuracy increases as the value of τ increases. The accuracy is not particularly sensitive to γ. For the three datasets, the accuracy of different methods varies within a small interval as γ increases.

For image segmentation, the effects of the parameter NS are also evaluated. As shown in Fig. 10, with the increasing number of superpixels from 50 to 200, the classification accuracy of the proposed methods also improves. When the number of training samples is small, the improvement is more pronounced.

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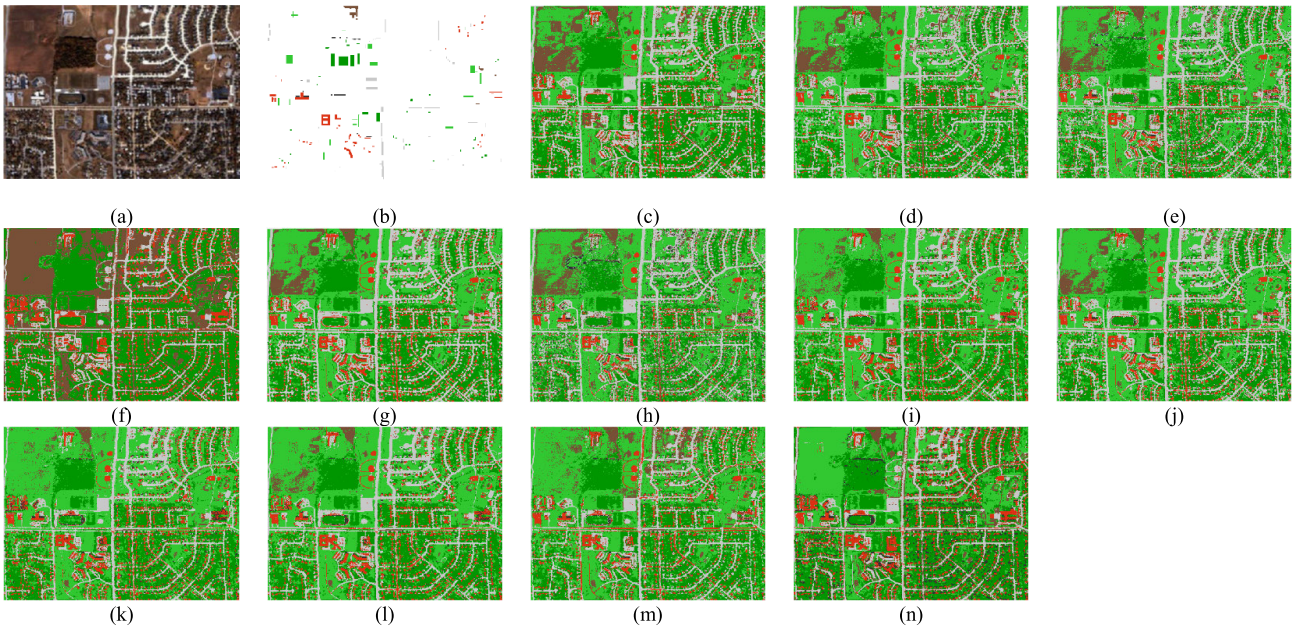


Fig. 12. Classification results of all samples from PD dataset. (a) False-color image. (b) Ground truth. (c) CR. (d) RCR. (e) CNN. (f) GAT. (g) GiGCN. (h) miniGCN. (i) GCN. (j) CR-GCN. (k) RCR-GCN. (l) SRCR. (m) SGCN. (n) SRCR-GCN.

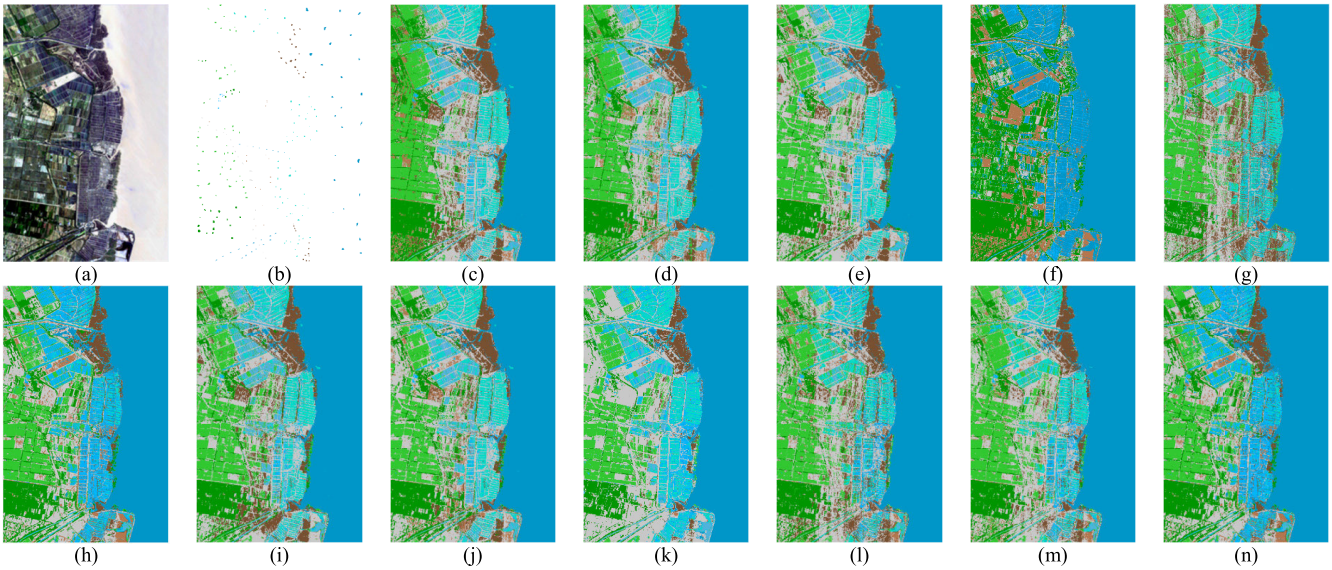


Fig. 13. Classification results of all samples from YC dataset. (a) False-color image. (b) Ground truth. (c) CR. (d) RCR. (e) CNN. (f) GAT. (g) GiGCN. (h) miniGCN. (i) GCN. (j) CR-GCN. (k) RCR-GCN. (l) SRCR. (m) SGCN. (n) SRCR-GCN.

TABLE IV
CLASSIFICATION ACCURACY FOR THE DC DATASET

Class	Train	CR	RCR	CNN	GAT	GiGCN	miniGCN	GCN	CR-GCN	RCR-GCN	SRCR	SGCN	SRCR-GCN
C1	10	53.58	92.00	84.31	42.69	89.54	94.98	99.25	98.83	98.51	96.96	97.51	**99.78**
C2	10	72.30	82.37	89.93	97.59	95.59	98.95	98.43	95.59	**100.00**	95.87	97.88	99.14
C3	10	100.00	96.55	99.83	100.00	98.96	98.10	89.81	97.75	**99.83**	77.66	99.65	98.39
C4	10	99.84	100.00	89.25	99.84	100.00	99.84	93.97	99.51	**100.00**	98.37	100.00	**100.00**
C5	10	97.16	89.54	99.40	98.06	94.92	98.36	98.80	99.10	99.55	97.73	99.55	**99.69**
C6	10	81.75	87.16	92.03	0.00	93.62	91.76	92.56	76.71	97.25	99.46	97.58	**99.73**
OA (%)		81.05	90.21	91.72	65.11	94.84	96.46	95.70	93.08	**99.00**	95.32	98.44	**99.49**
AA (%)		84.10	91.27	92.46	73.03	95.44	97.00	95.47	94.58	99.19	94.34	98.69	99.46
Kappa		0.7714	0.8814	0.8997	0.5893	0.9374	0.9570	0.9477	0.9164	0.9878	0.9430	0.9811	0.9939

3) Performance: It can be seen from Fig. 8 that the proposed methods obtain higher accuracy. Compared with SRCR, SRCR-GCN improves the accuracy by 5%∼10%. Figs. 11–13 separately show the classification results of different methods in the three experimental datasets.

For the first dataset, Table IV shows the classification results in the case of optimal parameters. Compared with

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TABLE V
CLASSIFICATION ACCURACY FOR THE PD DATASET

Class	Train	CR	RCR	CNN	GAT	GiGCN	miniGCN	GCN	CR-GCN	RCR-GCN	SRCR	SGCN	SRCR-GCN
C1	10	92.33	90.29	96.32	94.75	91.86	93.03	95.38	92.64	97.02	100.00	90.05	**100.00**
C2	10	79.62	88.41	79.26	51.72	90.31	93.93	97.46	76.99	97.01	90.59	98.46	**100.00**
C3	10	0.48	28.23	97.61	98.56	65.55	64.59	97.61	99.04	**97.13**	99.53	70.81	74.50
C4	10	99.73	100.00	100.00	97.56	97.56	100.00	98.64	100.00	98.37	100.00	94.85	**100.00**
C5	10	99.93	98.06	99.48	86.65	91.42	97.91	99.55	100.00	99.70	99.40	99.63	**100.00**
C6	10	41.80	75.92	79.53	24.00	78.27	84.00	81.25	77.02	91.37	69.38	92.94	**100.00**
OA (%)		77.13	86.82	90.15	68.43	87.73	91.71	93.87	88.47	**96.47**	90.97	94.28	**99.04**
AA (%)		68.98	80.15	92.03	75.54	85.83	88.91	94.98	90.95	96.77	93.15	91.12	95.75
Kappa		0.7141	0.8341	0.8761	0.6105	0.8462	0.8955	0.9229	0.8561	0.9555	0.8869	0.9278	0.9880

TABLE VI
CLASSIFICATION ACCURACY FOR THE YC DATASET

Class	Train	CR	RCR	CNN	GAT	GiGCN	miniGCN	GCN	CR-GCN	RCR-GCN	SRCR	SGCN	SRCR-GCN
C1	10	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00**	100.00	100.00	**100.00**
C2	10	85.52	94.21	88.80	98.65	92.08	95.95	99.42	89.19	**100.00**	100.00	100.00	**100.00**
C3	10	63.57	90.00	97.14	40.00	90.71	97.14	100.00	100.00	**100.00**	92.86	88.73	**100.00**
C4	10	100.00	98.73	94.94	89.87	93.67	100.00	100.00	100.00	98.73	92.41	100.00	**100.00**
C5	10	78.52	83.26	83.59	39.76	78.41	86.45	90.75	89.87	**99.12**	90.75	90.58	98.45
C6	10	31.58	40.23	88.35	82.71	42.86	53.38	80.83	25.56	89.85	86.47	84.64	**97.00**
C7	10	97.20	90.76	80.81	95.94	84.45	89.78	81.79	97.48	96.92	82.35	100.00	**100.00**
C8	10	37.64	73.10	88.59	61.41	84.51	79.35	90.90	99.73	99.05	90.35	95.76	**100.00**
C9	10	94.20	95.65	97.10	85.51	92.75	97.10	92.75	30.43	**100.00**	91.30	98.70	**100.00**
OA (%)		77.79	85.97	89.44	77.14	85.65	88.76	92.34	90.68	**98.54**	92.36	96.07	**99.50**
AA (%)		76.47	85.10	91.04	77.09	84.38	88.79	92.94	81.36	98.19	91.83	95.38	99.49
Kappa		0.7388	0.8332	0.8752	0.7324	0.8291	0.8664	0.9089	0.8881	0.9825	0.9092	0.9532	0.9941

TABLE VII
ABLATION ANALYSIS

Algorithm Description	GCN	√	√	√	√
	RCR		√		√
	Superpixel Segmentation			√	√
Datasets	DC	95.70	99.00	98.44	**99.49**
	PD	93.87	96.47	94.28	**99.04**
	YC	92.34	98.54	96.07	**99.50**

other methods, the accuracy of RCR-GCN and SRCR-GCN are obviously better, especially using small training set. The SRCR-GCN method can produce the best performance with 99.94% classification accuracy.

For the second dataset, SRCR-GCN improves the classification accuracy more significantly compared with the other two datasets. SRCR-GCN can improve the OA by 4.76% compared with SGCN and 2.57% compared with RCR-GCN. In addition, the accuracy of SRCR-GCN achieves 100% for all types except one. Table V shows the details.

For the third dataset, as shown in Table VI, the OA obtained by RCR-GCN can be improved by more than 10% compared with RCR and more than 5% compared with GCN. SRCR-GCN can increase further, its best accuracy is 99.50%, and it has a great capability of identifying the ground materials for the AA is more than 99%.

4) Ablation Analysis: From Table VII, ablation analyses show the superiority of the proposed method. On the basis of GCN, different improvement methods affected the classification performance in distinct degree.

First, for the dc dataset, the introduction of RCR resulted in a 3.30% improvement in accuracy, while for the PD dataset and YC dataset, it increased the accuracy by 2.60% and 6.20%, respectively. These results indicate that the introduction of RCR enhances the model's utilization of features, thereby improving its performance.

Second, the graph is constructed from the superpixel-based perspective and for the dc dataset, a 2.74% improvement was achieved, while for the PD dataset it increased by 0.41% and for YC dataset, it improved by 3.73%. These demonstrate the effectiveness of superpixel segmentation, especially for the dc and YC datasets.

Finally, both RCR and superpixel segmentation are introduced into GCN. For the dc dataset, it increased by 3.74%, for the PD dataset by 5.17%, and for the YC dataset by 7.16%. In summary, the experimental results clearly demonstrate the effectiveness of the proposed method.

V. CONCLUSION

In this article, a new method that combines RCR and GCN is proposed. Primarily, the representation coefficient of RCR is calculated to regarded as the graph edge weight which indicates the connection between nodes. Simultaneously, the diversity between features is well measured. Then, GCN is adopted to process graph data and explore graph structure. Due to the large amount of computation required for GCN, SLIC is further employed to reduce the amount of calculation that the graph is performing on superpixel nodes. The experimental results demonstrate that both of our methods can effectively improve the classification performance compared with other methods.

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Effective utilization of the edge weights of the graph structure can indeed improve the classification accuracy, but the computational cost cannot be ignored. In the future, it may need to explore a more efficient approach to construct the graph for HSI classification.

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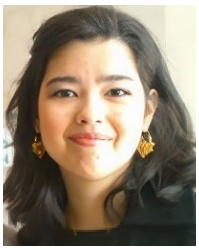
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